Project Proposal: Mini Wheeled Swarm Robotics

1. General Idea

Build a group of 3 to 4 small, autonomous wheeled car rovers. Each robot can make its own decisions using its onboard brain, but they communicate with each other to complete a shared task (like moving objects together) without crashing.

2. Project Objectives

- **2D Movement:** Build rovers that can move, turn, and navigate precisely on a flat surface (X, Y coordinates).
- **Self-Localization:** Every car must know exactly where it is located on the map at any given second.
- **Swarm Awareness:** Every car must know where its neighbor cars are located to coordinate movements.
- **Target Navigation:** The cars must navigate toward a target goal location, execute a task, and move to an end location.
- **Collision Avoidance:** The cars must sense their immediate surroundings to avoid hitting walls or each other.

3. Hardware Architecture (The Build List)

- **Brain: ESP32 Development Board.** Handles calculations and uses its built-in Wi-Fi/Bluetooth for wireless communication.
- **Motors: DC Geared Motors** paired with an **H-Bridge Motor Driver** (like the L298N or TB6612FNG) to control speed and direction.
- 🚨 **Missing Piece Added — Custom Feedback: DIY Magnetic Encoders.** Attaching small magnets to the wheels and using Hall-Effect sensors (like the US1881) connected to the ESP32 to count wheel spins. This is called *Odometry*—it helps the car calculate how many centimeters it traveled between camera updates.
- **Global Eye: Overhead Camera.** Mounted above the driving arena. It looks down, tracks visual tags (like ArUco markers or AprilTags) printed on top of each car, and calculates everyone's position.

- **Tool (End Effector): Electromagnet.** Connected to a relay or transistor so the ESP32 can turn it ON to pick up a metal object and OFF to drop it.
- 🚨 **Missing Piece Added — Local Safety: ToF (Time-of-Flight) Distance Sensors.** The overhead camera is good for global coordinates, but it has a tiny lag. You need at least 1 or 2 small distance sensors (like the VL53L0X) on the front of each car as an emergency brake so they never physically dent each other.

4. Missing Core Challenges (Add these to your Word file!)

You have a great foundation, but you need to add sections addressing these 3 specific swarm problems:

A. The Communication Protocol

- **The Problem:** Standard Wi-Fi routers introduce lag when 4 devices try to talk at the exact same time.
- **The Solution to Add:** Use **ESP-NOW**. This is a built-in feature of the ESP32 that lets the cars flash data directly to one another peer-to-peer in milliseconds without needing a Wi-Fi router.

B. The Overhead Camera System

- **The Problem:** A normal camera stream cannot be read directly by an ESP32 because video processing takes too much memory.
- **The Solution to Add:** The overhead camera must plug into a laptop running **Python + OpenCV**. The laptop reads the video, finds the cars using print-out markers, and sends just the simple text coordinates (X, Y) back to the ESP32s over the network.

C. Task Allocation (Who does what?)

- **The Problem:** If there is one object to pick up, how do the cars decide which *one* goes to get it? If they all rush at it, they will jam.
- **The Solution to Add:** Implement a basic rule-set. For example: *"The car closest to the target becomes the Leader and goes for the pickup; the other cars move 20 cm away to clear a path."*